PO11Q - Quantitative Political Analysis:
From Measurement to Inference

Week 10, Exercise Solutions

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Exercises

1. A researcher claims that their study has 75% power to detect the effect they are interested in.
   a) This means that if the effect really exists, the study has a 75% chance of producing a statistically significant result.
   b) If the study were repeated 100 times and the effect were real, it would fail to reach significance about 25 times. This is called a Type II error (or false negative).
   c) The four factors that determine statistical power are:
      - Effect size: Larger effects increase power because they are easier to distinguish from random variation.
      - Sample size: Larger samples increase power because they reduce sampling variability.
      - Significance level (α): Larger values of α increase power because it is easier to reject the null hypothesis.
      - Population variability (σ): Greater variability lowers power because observations are more spread out.
   d) Larger sample sizes reduce sampling variability, making sample means cluster more closely around the population mean. As a result, true effects are easier to distinguish from random fluctuations, increasing power.
2. A study reports Cohen's $d = 0.35$ for the effect of a new teaching method on exam scores.
   a) According to Cohen's conventions, this would be considered a small-to-medium effect (typically classified as small).
   b) The raw difference in means is
   $$d \cdot s = 0.35 \cdot 15 = 5.25.$$
   The treatment and control groups differ by approximately 5.25 points.
   c) Smaller effects produce less separation between the null and alternative distributions, making them harder to distinguish from sampling error. Therefore, larger samples are required to detect them reliably.
   d) Because the effect is relatively small, a reasonably large sample size would be needed to achieve adequate power.

3. A researcher tests whether meditation reduces stress levels. Without meditation, the population mean stress score is 65. A sample of 40 people who practiced meditation for 8 weeks had a mean stress score of $\bar{y} = 59$ with sample standard deviation $s = 11$.

   a) The observed effect size is

   $$d_{\text{obs}} = \frac{\bar{y} - \mu_0}{s} = \frac{59 - 65}{11} = -0.545.$$

   b) The magnitude is approximately $|d| = 0.55$, which is typically interpreted as a medium effect.

   c) The test statistic is

   $$t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}} = \frac{-6}{11/\sqrt{40}} \approx -3.45.$$

   d) Using

   $$t = d_{\text{obs}}\sqrt{n},$$

   we obtain

   $$(-0.545)\sqrt{40} \approx -3.45,$$

   which matches the value from part (c).

4. You are planning a study to test whether a new app improves sleep quality. Previous research suggests a medium effect size ($d_{\text{test}} = 0.5$). You want 80% power with $\alpha = 0.05$ (two-tailed).

   a) Using the lecture approximation,

   $$n \approx \left(\frac{2.8}{0.5}\right)^2 \approx 31.4,$$

   so approximately 32 participants are required.

   b) The noncentrality parameter is

   $$\lambda = d_{\text{test}}\sqrt{n} = 0.5\sqrt{32} \approx 2.83.$$

   c) The noncentrality parameter measures how far the alternative distribution is shifted away from the null distribution. Larger values imply greater separation and therefore greater power.

   d) No. Since

   $$\lambda = d\sqrt{n},$$

   doubling n multiplies λ by $\sqrt{2} \approx 1.41$, not by 2.

   e) With only 20 participants, power would be lower than 80%, making it more likely that a real effect would go undetected.

5. A researcher plans a one-sample test, $H_0 : \mu = 100$ versus $H_1 : \mu \neq 100$, with $\alpha = 0.05$.

   a) Under the null hypothesis, the central t distribution is centred at 0.

   b) A large noncentrality parameter means the alternative distribution is further from the null distribution. This reduces overlap and increases power.

   c) Power is the probability of rejecting H_0 when the alternative hypothesis is true. To calculate this probability, we must know the distribution of the test statistic under the alternative hypothesis. The central distribution describes only what happens when H_0 is true and therefore cannot be used to calculate power by itself.

6. A researcher conducts a study with $n = 20$ to test whether a new therapy reduces anxiety ($H_0 : \mu = 50$). They find $\bar{y} = 46$, $s = 10$, $d_{\text{obs}} = 0.4$, and $p \approx 0.09$.

   a) The study may have had insufficient power. The observed effect is non-trivial, but the sample size may have been too small to detect it reliably.

   b) The conclusion is not justified. A non-significant result does not prove that the effect is zero; it simply indicates insufficient evidence to reject the null hypothesis.

   c) The researcher should consider collecting a larger sample, conducting a formal power analysis, and examining confidence intervals around the estimated effect.

7. You are designing a study to test whether a mindfulness intervention reduces burnout among teachers. A colleague's pilot study ($n = 15$) found $\bar{y} = 38$, $s = 10$, while the population mean is $\mu_0 = 45$.

   a) The observed effect size is

$$d_{\text{obs}} = \frac{38 - 45}{10} = -0.7.$$

   b) No. Pilot studies often produce unstable estimates because they are based on small samples. It is usually safer to plan using a somewhat smaller, more conservative value.

   c) Example R code:

```r
power.t.test(
  delta = 0.4,
  sd = 1,
  sig.level = 0.05,
  power = 0.80,
  type = "one.sample",
  alternative = "two.sided"
)
```

   d) Example R code:

```r
power.t.test(
  n = 25,
  delta = 0.4,
  sd = 1,
  sig.level = 0.05,
  type = "one.sample",
  alternative = "two.sided"
)
```

e) Possible options include:

- Recruit more participants.
  - Advantage: increases power directly.
  - Disadvantage: may be expensive or impractical.
- Accept a lower level of power.
  - Advantage: study can proceed with available resources.
  - Disadvantage: greater risk of Type II errors.
- Use a more precise design (e.g., reduce measurement error).
  - Advantage: can increase power without increasing sample size.
  - Disadvantage: may require additional resources or redesign.
- Use a one-tailed test if strongly justified in advance.
  - Advantage: increases power.
  - Disadvantage: inappropriate if effects in the opposite direction are plausible.